

EBU6505 – Reasoning and Agents

Block 2, Day 5 – Dynamic Decision Networks and POMDP Synthesis

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Teaching Week 4

What have we covered this week?

- Day 1: Bayesian networks for static uncertain reasoning
- Day 2: exact inference in Bayesian networks
- Day 3: temporal models such as hidden Markov models (HMMs) and dynamic Bayesian networks (DBNs)
- Day 4: planning with uncertainty using utility and Markov decision process (MDP) ideas

Main idea: This week moved from uncertain belief, to temporal belief, and then to action under uncertainty.

What are we going to cover today?

- how a dynamic decision network (DDN) adds decisions and utilities to a temporal graph
- what a partially observable Markov decision process (POMDP) adds when the true state is hidden
- how belief updates support action selection
- how to choose between a Bayesian network, dynamic Bayesian network, Markov decision process, dynamic decision network, and POMDP
- one simple healthcare case plus other domain examples

Main idea: Today is a synthesis lecture. The goal is to connect all the models, not to add many new symbols.

- *Artificial Intelligence: A Modern Approach*, 4th ed.
Chapter 17: Making Complex Decisions
Chapter 18: Probabilistic Reasoning over Time
- *Artificial Intelligence: Foundations of Computational Agents*, 3rd ed.
Chapter 12: Planning with Uncertainty
Chapter 13: Probabilistic Inference

Main idea: Use the AIMA reading for dynamic decision network and POMDP language, and the Poole reading for planning under uncertainty.

Learning Outcomes for Today

By the end of this session, you should be able to:

- explain how a dynamic decision network (DDN) extends a temporal probabilistic model with decisions and utilities
- explain the parts of a partially observable Markov decision process (POMDP) in simple language
- describe how action, observation, and belief fit together
- choose a suitable model for a simple uncertain decision problem
- explain one end-to-end example from sensing to action

Why Do We Need One Synthesis Model?

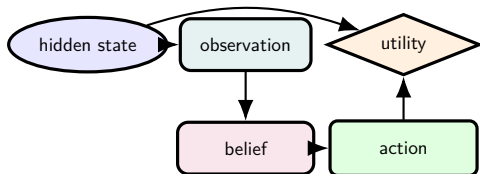
Question: Why are Bayesian networks, dynamic Bayesian networks, Markov decision processes, and POMDPs usually taught together at the end?

What we already have

- uncertain facts
- uncertain facts over time
- actions and utilities

What we still need

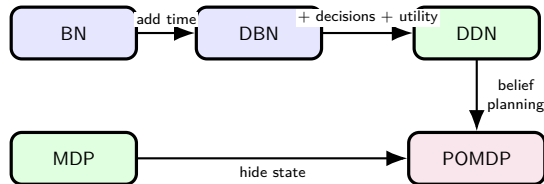
- one model for hidden state, observations, actions, and utility



Main idea: Day 5 joins hidden state, observations, actions, and utility into one model.

Concept Map: How the Models Connect

Question: How do Bayesian networks, DBNs, MDPs, DDNs, and POMDPs connect?



Simple reading

BNs and DBNs model belief.

Simple reading

MDPs, DDNs, and POMDPs add action.

Main idea: DDNs and POMDPs bring time, uncertainty, and action together.

What Is a Dynamic Decision Network?

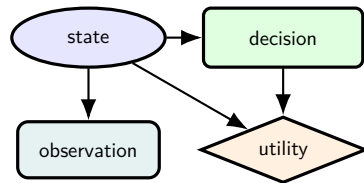
Question: What does a dynamic decision network add to an ordinary temporal probabilistic graph?

DDN idea

Start from a DBN.
Keep the time slices.
Add one decision node and one utility node.

Why useful?

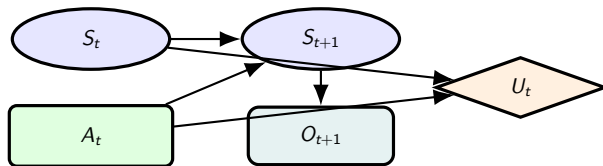
Now the model says both what may happen
and which action is better.



Main idea: A DDN is a temporal uncertainty model with choices and consequences added.

Two-Slice DDN Template

Question: What does one step of a DDN usually look like?



Symbols used on this slide

S_t : state now, A_t : action now, S_{t+1} : next state

O_{t+1} : next observation, U_t : utility for this choice

Main idea: A dynamic decision network shows how current state and action affect the next state, the next observation, and the utility.

What Do the DDN Arrows Mean?

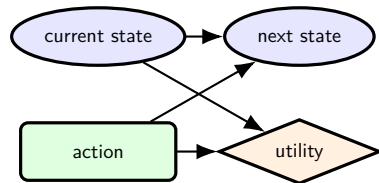
Question: What kind of influence does each arrow in a DDN represent?

Arrow meanings

state $\rightarrow$ next state: world dynamics

action $\rightarrow$ next state: the choice changes what may happen

state + action $\rightarrow$ utility: reward depends on both



Main idea: The arrows show what the choice depends on and what the choice can change.

From DBN to DDN

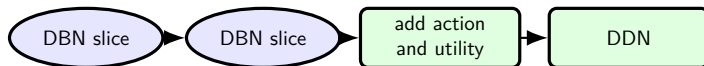
Question: How do we move from temporal belief tracking to temporal decision-making?

DBN

hidden state over time
noisy observations
no explicit decision or utility

DDN

hidden state over time
noisy observations
plus decisions and utilities



Main idea: A dynamic decision network is a dynamic Bayesian network plus decision-making structure.

What Is a POMDP?

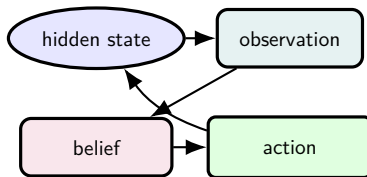
Question: What do we need when the true state is hidden but the agent must still act?

Main challenge

the agent cannot directly see the true state

Main solution

the agent plans using a belief over states instead of one known state



Main idea: A partially observable Markov decision process (POMDP) is the standard model for sequential decisions when the world state is partly hidden.

The POMDP Pieces

Question: What are the main pieces of a POMDP?

$$\mathcal{P} = (S, A, O, T, Z, R, \gamma)$$

State and action

S : hidden states
 A : actions
 O : observations

Models and reward

T : transition model
 Z : observation model
 R : reward, γ : discount

Main idea: A POMDP says how the world changes, what we can observe, and which actions are better.

Belief Update Intuition

Question: How does the agent change its belief after acting and observing?

$$b_{t+1}(s') = \eta Z(o_{t+1} \mid s', a_t) \sum_s T(s, a_t, s') b_t(s)$$

Step 1: predict

use the transition model to predict the next hidden state

Step 2: correct

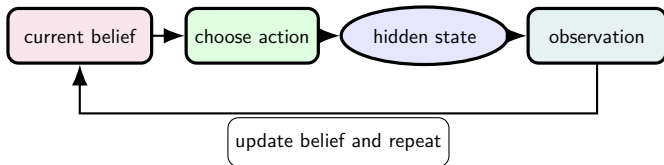
use the new observation to correct that prediction

Here, $b_t(s)$ is the current belief and η rescales the updated numbers so they sum to 1.

Main idea: Belief update means: predict first, then correct with new evidence.

Action-Observation-Belief Loop

Question: What is the repeating loop inside a POMDP?



Main idea: The agent acts, receives observations, updates belief, and then chooses again.

Mini Belief-State Example

Question: How can one observation change the best action?

Before warning signal

low risk: 70%
high risk: 30%
best action: continue monitoring

After warning signal

low risk: 35%
high risk: 65%
best action: alert clinician



belief before signal



belief after signal

Main idea: In a POMDP, one new observation can change the belief and therefore change the best action.

DDN Versus POMDP

Question: What is the main difference between a DDN and a POMDP?

feature	DDN	POMDP
main view	graphical decision model over time	sequential planning model with hidden state
what it emphasizes	nodes and dependencies	belief update and action choice
best classroom use	showing structure clearly	showing the action-belief loop clearly

Main idea: DDN and POMDP are closely related. A dynamic decision network highlights the graph structure, while a POMDP highlights belief-state planning.

One Hospital Robot Story

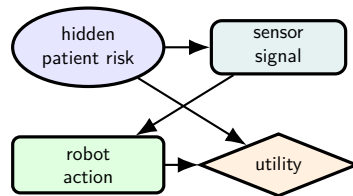
Question: How do DDN and POMDP ideas appear in one simple healthcare example?

One simple story

hidden part: patient risk

evidence: noisy monitor signals

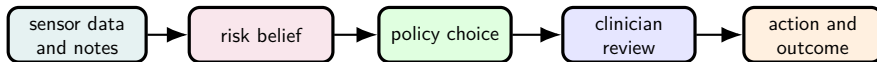
action: continue, ask for help, or alert a clinician



Main idea: The same story can be read as a DDN graph or as a POMDP planning problem.

Healthcare Decision-Support Pipeline

Question: How does the full reasoning pipeline run from data to action?



Before action

data is used to estimate belief

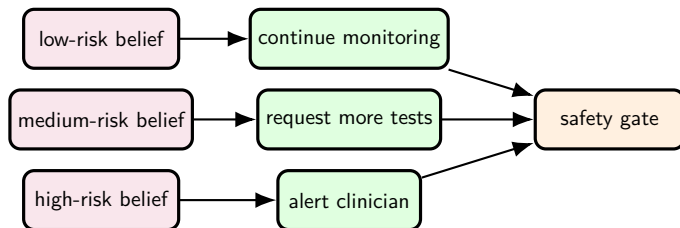
At action time

policy and safety checks decide whether to continue, ask for review, or escalate

Main idea: The model does not act on raw data directly. It acts on a belief that has already been updated.

Belief-State Policy with Safety Gates

Question: How can one policy change its action as the risk belief changes?



Main idea: Belief guides the action, but a safety gate can still block or confirm risky choices.

What Can Go Wrong?

Question: Why can a DDN or POMDP system still make poor decisions?

Common problems

- wrong transition model
- wrong observation model
- wrong utility design
- missing context

Why it matters

the system may look mathematically correct but still choose harmful or costly actions

Main idea: A good algorithm is not enough. The model and the utility must also match the real task.

How Do We Reduce Risk?

Question: What simple checks make a DDN or POMDP system safer?

Model checks

- calibrate sensors
- validate transition assumptions
- test utility trade-offs

Decision checks

- keep human approval for costly actions
- keep a safe fallback action
- ask for more evidence when belief is uncertain

Main idea: Safer decision-making comes from both better models and better control rules.

Which Model Should I Use?

Question: How do I choose between BN, DBN, MDP, DDN, and POMDP?

model	use when	key limit
BN	one-time uncertain reasoning	no time and no decisions
DBN	temporal belief tracking	no explicit utility or action choice
MDP	sequential decisions with visible state	breaks when state is hidden
DDN	temporal decision graph with utility	still needs a clear state model
POMDP	sequential decisions with hidden state	harder to solve computationally

Main idea: Choose the smallest model that still matches the task structure.

Cross-Domain Map

Question: Where do DDN and POMDP ideas appear outside healthcare?

Autonomous driving

hidden driver intent; noisy sensing; safety actions

Cybersecurity

hidden attack stage; noisy alerts; staged response

Financial risk

hidden fraud risk; partial signals; intervention choices

Transport and energy

hidden demand; uncertain signals; sequential control

Main idea: The same pattern appears again and again: hidden state, noisy evidence, and repeated choices.

Example: Autonomous Driving

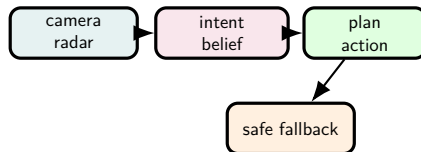
Question: How does hidden-state planning appear in self-driving systems?

Hidden part

other road users may brake, turn, or cross

Decision part

the car may slow down, continue, or choose a safer maneuver



Main idea: Driving needs both prediction and safe action under uncertainty.

Example: Finance and Cybersecurity

Question: What do finance and cybersecurity have in common with POMDP thinking?

Finance

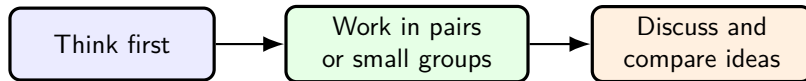
hidden fraud risk
partial account signals
actions such as allow, flag, or freeze

Cybersecurity

hidden attack stage
noisy alert streams
actions such as monitor, isolate, or escalate

Main idea: Both domains require repeated action under hidden risk, not just one-time classification.

In-Class Exercises



Use this time to work on today's in-class tasks and prepare one clear answer you can discuss.

Main idea: Use the exercise time to connect hidden state, evidence, belief, action, and utility in one clear explanation.

Summary

- a DDN adds decisions and utilities to a temporal probabilistic graph
- a POMDP adds hidden state, observations, and belief-state planning
- belief update is the bridge between sensing and action
- the right model depends on time, observability, and decision needs
- the same pattern appears in healthcare, driving, finance, and cybersecurity

Main idea: Day 5 closes the block by connecting uncertain reasoning, temporal reasoning, and decision-making.

Block 2 Close-Out

- belief under uncertainty: Bayesian networks
- temporal belief: HMMs and DBNs
- visible-state planning: MDP ideas
- hidden-state planning: DDN and POMDP synthesis
- next step: apply the right model to your own scenario

Main idea: Block 2 gives you one complete arc from uncertain evidence to informed action.

Further Study

- Read Poole Chapters 12–13 and revisit AIMA Chapters 17–18.
- Practice: Poole Ch.12, Exercises 12.13, 12.15, 12.16, 12.17, and 12.18.
- AIMA complex-decisions exercise page: 17.15, 17.16, and 17.17.
- AIMA probabilistic-reasoning-over-time exercise page: 15.15, 15.17, and 15.18.
- Focus on belief-state update, partially observed action choice, and how hidden-state planning differs from standard Markov decision process planning.